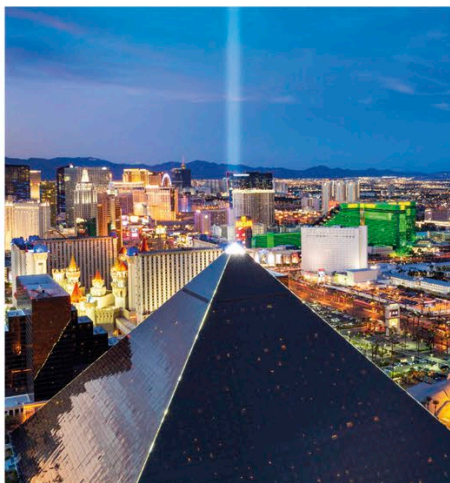


Energy-efficient lights

- **What?** Fluorescent lights
- **Who?** Edmund Germer, Friedrich Meyer, and Hans Spanner
- **Where and when?** Germany, 1926

In a fluorescent light, electricity is passed through vapor of the element mercury, creating invisible ultraviolet light. When this hits a fluorescent coating lining the inside of the bulb, it is converted to visible light. Early fluorescent lights were large, but in the 1970s, small compact fluorescent lights (CFLs) were developed by the American engineer Edward E Hammer.



Searchlights

- **What?** Xenon arc lamp
- **Who?** Osram Licht AG
- **Where and when?** Germany, 1940s

A type of arc light, a xenon lamp is filled with high-pressure xenon gas. When electricity passes through it, the xenon glows brightly. Xenon lamps are used in movie projectors, searchlights, lighthouses, and to create the most powerful light beam in the world—on top of the Luxor Hotel in Las Vegas (above).



CFLs last up to 15 times longer than incandescent light bulbs.

Laser light

- **What?** Laser
- **Who?** Theodore H. Maiman, Arthur Schawlow, Charles Townes, and Gordon Gould
- **Where and when?** US, 1960

A laser is a powerful but very narrow beam of light. It is produced by providing energy to the atoms of a solid or gas so they give off light. The atoms work in step with each other to produce very pure light. Lasers are mainly used for industrial purposes, such as carrying information along fiber-optic cables and sending information into space.



Lasers can be used to spectacular effect at concerts

LEDs

- **What?** Light-emitting diodes (LEDs)
- **Who?** Nick Holonyak, Jr.
- **Where and when?** US, 1962

Because they use less energy, last longer, and are less fragile, LEDs have begun to replace incandescent bulbs in recent times. LEDs work by jiggling the electrons in a material known as a semiconductor, such as silicon, so that an electric current flows, creating light. LED bulbs can be very small—they are used for the tiny lights seen on power buttons on remote controls.

Colored decorative lights are often LEDs.

